



PTT PHENOL COMPANY
LIMITED

PROCESS SPECIFICATION

98% SULFURIC ACID INJECTION TANK (D-2310)

Docu. No. : 14780-8120-PS-D2310
Project No. : 120117
Rev. No. : Z1
Date : MAY 10, 2016
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TITLE :

98% SULFURIC ACID INJECTION TANK

(D-2310)

DISCIPLINE : PROCESS
DOCUMENT NO. : 14780-8120-PS-D2310
PROJECT NAME : PTT Phenol Train II Project
PROJECT NO. : 120117
OWNER : PTT Phenol Company Limited (PPCL)

Z1	MAY 10, 2016	AS BUILT	S.W.Park	J.S.Chung	-	S.W.Lee
F1	JAN 19, 2015	FINAL ISSUE	S.W.Park	J.S.Chung	-	S.W.Lee
F1	JAN 19, 2015	FINAL ISSUE	S.W.Park	J.S.Chung	-	S.W.Lee
A1	JAN 09, 2015	APPROVED FOR DESIGN	S.W.Park	J.S.Chung	-	S.W.Lee
O1	NOV 18, 2014	COMMENT ISSUE	S.W.Park	J.S.Chung	-	S.W.Lee
REV. NO.	DATE	DESCRIPTION	PREP'N	REVIEW	QA REVIEW	APPROVAL
REV. NO.	PREPARATION DEPT.	PREPARATION	REVIEW	QA REVIEW	APPROVAL	
Z1	PROCESS	S.W.Park	J.S.Chung	-	S.W.Lee	
		MAY 10, 2016	MAY 10, 2016	-	MAY 10, 2016	

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Revision
Indication



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VESSELS

PROJECT SPECIFICATION

963766 — 301

SHEET 3 / 4

REV	DATE	BY	APVD	REV	DATE	BY	APVD
0	29MAR12	JG	MYW				

General Notes

- A. Vessels shall be constructed in accordance with the following unless otherwise specified on the individual vessel specification sheet.

Standards:

UOP Standard Specification 3-11
UOP Standard Drawing 8-121

- B. Although the resulting inside diameter may be slightly different than specified, vessels without trays or packing, may be fabricated from pipe.
- C. The dimensioned location of some of the vessel level nozzles may be noted as min (minimum) or max (maximum). It is the responsibility of the contractor to adjust these dimensions to suit the specific level requirements specified on the UOP Piping and Instrument Diagram(s) and on UOP Standard Drawing 8-121. When additional nozzles are required for level instrumentation, the contractor shall specify them consistent with the size, class and facing requirements specified in the UOP Project Specification 301-Vessels and on the UOP Piping and Instrument Diagram(s).
- D. When system of measurement is metric, all dimensions are in millimeters unless noted otherwise.
- E. Contractor shall not locate TI thermowells in a downcomer. Contractor shall also ensure that TI thermowells are located to avoid interference with downcomers, weirs, distributors, collectors, baffles, chimneys, traps and all other vessel internals.

Other specifications
referenced in this
specification:

UOP Standard Specification: 3-11, 3-15

UOP Standard Drawing: 3-115, 3-122, 3-123, 3-183, 3-190, 3-272, 3-278, 3-280, 3-305, 3-313, 3-322, 8-121

UOP Project Specification: 963766-307, 963766-401, 963766-808

POSCO ENGINEERING

VESSEL DATA SHEET

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CLIENT : PTT PHENOL JOB NAME : PTT PHENOL TRAIN II PROJEC T JOB NO : 120117 PLANT LOCATION : MAP TA PHUT, THAILAND REV DOC. NO : 14780-8120-PS-D2310					REV.	DATE	REV.	DATE
					O1	NOV 18, 2014		
					A1	JAN 09, 2015		
					F1	JAN 19, 2015		
					Z1	MAY 10, 2016		

Item No. D-2310	Service 98% SULFURIC ACID INJECTION TANK	1
No. Req'd 1 (One)	Type Vertical	2
Regulation	Code ASME SECTION VIII DIV.1	3

DESIGN CONDITIONS			
Fluid Name	98% SULFURIC ACID		
Density at Oper. Temp. kg/m ³	1728.7		
Shell Inside Diameter mm	1000		
Pressure	Oper.	0.01	
kg/cm ² g	Design	3.5	
Temperature	Oper.	40	
°C	Design	120	
	MDMT	15	
Material	Shell	SA 240 TYPE 304 L	
	Head	SA 240 TYPE 304 L	
	INTL		
	Support	SA 240 TYPE 304 L	
Corrosion	Shell	1.5	
Allowance	Head	1.5	
mm	INTL		
PWHT	MR		
Radiograph	MR		
Joint Efficiency			
This Vessel is in the Following Service(s):			
Hydrogen	NO		
Hydrofluoric (HF) Acid	NO		
Hydrofluoric (HF) Acid (with Resid. Elem. Control)	NO		
Wet Hydrogen Sulfide (H ₂ S)	NO		
Cyclic	NO		
Caustic	NO		
Carbonate	NO		
Amine	NO		
Vessel Heads :	2:1 ELLIPTICAL TOP/BOTTOM		
Insulation :	NO		
Weight :	Empty	,Full	

NOZZLE SCHEDULE				
Mark	Qty.	Size	Rating & Facing	Service
MO1	1	8	CL 150, RF	INSPECTION
EO1	1	2	CL 150, RF	EQUALIZING
IO1	1	2	CL 150, RF	INLET
RO1	1	2(note2)	CL 150, RF	CONSERVATION VENT
VO1	1	2	CL 150, RF	VENT
WO1	1	2	CL 150, RF	OUTLET
LO1	1	2	CL 300, RF	LEVEL
LO2	1	2	CL 300, RF	LEVEL

ACCESSORIES	

Note

1. READYMADE TOTE IS ACCEPTABLE WITH 0.55 M3 OF OPERATING VOLUME. (OPERATING RANGE OF LIQUID LEVEL : 25 % TO 75 %)

2. INLET LINE SIZE OF CONSERVATION VENT TO BE CONFIRMED BY CONSERVATION VENT VENDOR

3. THE LOCATION OF L01 NOZZLE CAN BE ON SIDE OR TOP OF THE VESSELE AS PER SELECTED LEVEL GAUGE. TO BE COMFIRMED LATER.

UNIT : mm

